TONNAGE AVAILABILITY AND FREIGHT RATE DYNAMICS IN THE VLCC MARKET

- A Study of Position List Indicators *

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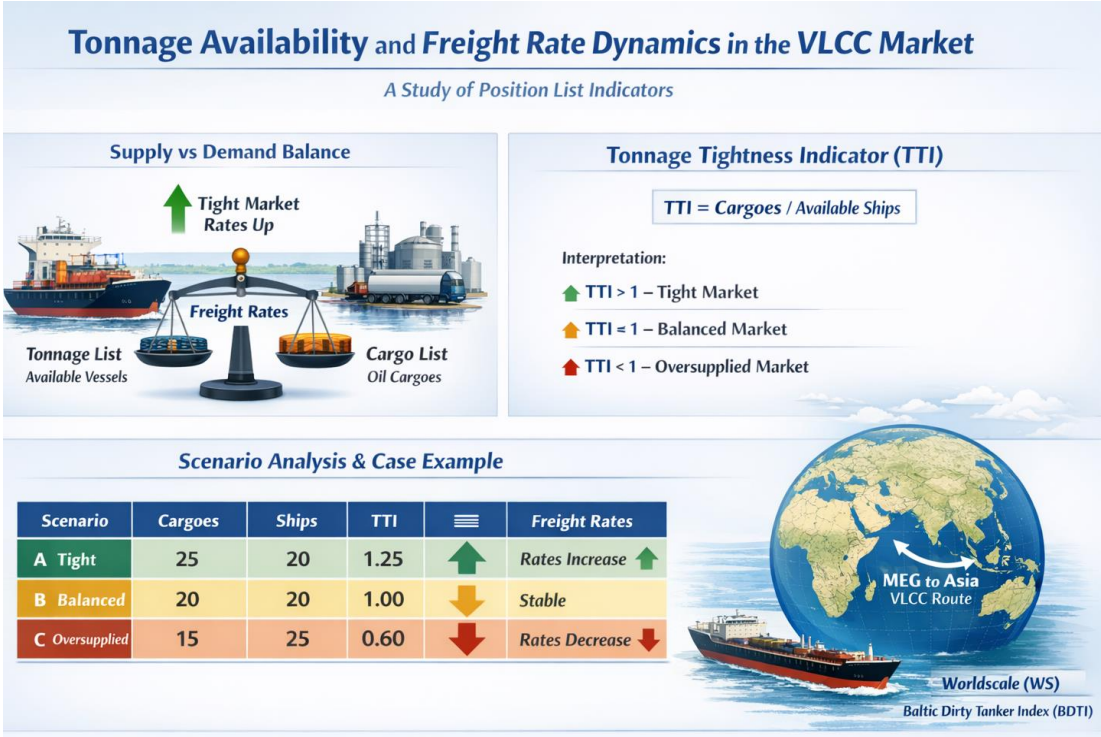
1. Introduction

The global tanker shipping market plays a critical role in facilitating crude oil trade, particularly between major exporting regions in the Middle East and key importing economies in Asia. This trade is primarily serviced by **Very Large Crude Carriers (VLCCs)**, which transport large volumes of crude oil across long-haul routes.

One of the most actively traded routes is from the **Middle East Gulf (MEG) to Asia**, especially China and India. Freight rates in this market are highly volatile and are influenced by a variety of factors. Among these, **tonnage availability**—the number of ships open for employment—remains one of the most fundamental drivers.

Shipbrokers operating in this market rely heavily on **position lists**, which provide real-time information on vessel availability. Understanding how these lists interact with cargo demand is essential to anticipating freight rate movements.



2. Research Objective

This study aims to analyse how tonnage availability in key loading regions influences short-term freight rate movements in the VLCC market.

3. Research Question and Hypothesis

Research Question
How does tonnage availability in the Middle East Gulf affect freight rate dynamics in the VLCC segment?

Hypothesis
A reduction in available VLCC tonnage leads to increased freight rates, while excess tonnage results in downward pressure on rates.

4. Conceptual Framework

The tanker freight market operates on a simple but highly sensitive mechanism:

- **Tonnage List (Supply):** Number of available vessels in a region
- **Cargo List (Demand):** Number of cargoes requiring shipment

Freight rates are determined by the interaction between these two forces.

- When cargo demand exceeds vessel supply, charterers compete for ships, pushing freight rates higher
- When vessel supply exceeds cargo demand, shipowners compete for cargoes, driving rates lower

A key characteristic of tanker markets is that even small changes in vessel availability can result in **disproportionate rate movements**, due to the relatively fixed size of the global fleet in the short term.

5. Analytical Model: Tonnage Tightness Indicator (TTI)

To quantify market conditions, this study introduces a simple indicator:

TTI = Amount of Cargoes / Number of Available Ships

Interpretation

- **TTI > 1** → Tight market → Upward pressure on freight rates
- **TTI ≈ 1** → Balanced market → Stable rates
- **TTI < 1** → Oversupplied market → Downward pressure on rates

This indicator provides a simplified yet effective way to assess short-term market conditions.

6. Scenario Analysis

To illustrate the relationship between tonnage availability and freight rates, three market scenarios are considered:

1. Tight Market

- Limited number of VLCCs available in MEG
- Strong cargo demand from Asian importers
- **Outcome:** Increased competition among charterers → Rising freight rates

2. Balanced Market

- Vessel supply closely matches cargo demand
- **Outcome:** Stable freight rates with limited volatility

3. Oversupplied Market

- Excess vessels open for employment
- Fewer cargoes in the market
- **Outcome:** Increased competition among shipowners → Declining freight rates

7. Mini Case Example (Applied Analysis)

To demonstrate the practical application of the TTI model, consider the following hypothetical scenarios in the **MEG–China VLCC market**:

Scenario	Cargoes	Available Ships	TTI	Market Condition	Freight Direction
A	25	20	1.25	Tight	Rates Increase
B	20	20	1.00	Balanced	Stable
C	15	25	0.60	Oversupplied	Rates Decrease

Interpretation

- In **Scenario A**, limited vessel availability creates upward pressure on freight rates
- In **Scenario B**, equilibrium leads to stable pricing
- In **Scenario C**, excess tonnage results in weaker rates

This example demonstrates how even a simple ratio can provide meaningful insight into market direction.

8. Industry Relevance

In real-world shipping markets, freight rates are commonly quoted using the **World scale (WS)** system, and broader market trends are tracked using indices such as the **Baltic Dirty Tanker Index (BDTI)**.

Shipbrokers continuously monitor:

- Vessel positions
- Cargo volumes
- Regional supply-demand imbalances

By interpreting position lists effectively, brokers can anticipate market movements and negotiate optimal fixtures for their clients.

9. Conclusion

This study highlights the critical role of tonnage availability in determining freight rate movements in the VLCC market. By examining the balance between vessel supply and cargo demand, it demonstrates how fundamental market dynamics drive freight volatility.

The introduction of the **Tonnage Tightness Indicator (TTI)** provides a simple yet practical framework for analysing market conditions. While the model is simplified, it reflects the core principles used by shipbrokers in real-world decision-making.

Understanding these dynamics is essential for participants in the tanker market, particularly in highly active trade routes such as the Middle East Gulf to Asia.

10. References

Baltic Exchange – Baltic Dirty Tanker Index (BDTI)

International Energy Agency (IEA) – Oil Market Data

UNCTAD – Review of Maritime Transport